Jay Sejin Sim, Ph.D.

NSF GRFP Fellow

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EDUCATION

Ph.D. in Mechanical Engineering October 2026

Stanford University, Stanford, CA
Advisor: Prof. Ruike Renee Zhao

Bachelor of Science in Mechanical Engineering with Honors May 2022

The Pennsylvania State University | Schreyer Honors College, University Park, PA
Dean's List: All Semesters

RESEARCH FOCUS

My research focuses on using mechanics to program the behavior of physical systems, with interests in active and architected materials, mechanical metamaterials, and stimuli-responsive structures. My long-term goal is to establish an independent research program investigating how the design of physical systems and their spatiotemporal excitation can be used to control mechanical, acoustic, thermal, optical, electromagnetic and other physical phenomena. I am committed to pursuing an academic career that integrates fundamental research with engineering education and the mentorship of future researchers.

HONORS AND AWARDS

Stanford Bio-X Poster Award, September 2025

Future Faculty Symposium Travel Award, Society of Engineering Science June 2025

Centennial Teaching Assistant Award, Stanford University May 2025

NSF Graduate Research Fellowship, National Science Foundation April 2024

Outstanding Undergraduate Research Achievement, PSU April 2022

2nd place in Engineering, PSU Undergraduate Research Exhibition April 2021

3rd place in Engineering, PSU Undergraduate Research Exhibition April 2020

PUBLICATIONS (* denotes equally contributing first authors)

1. **Sim, J.**, Leanza, S., Hwang, S., Ogaki, C., & Zhao, R. R., Twist-Programmable Rod Origami (RodOri) Metamaterials: Harnessing Cellular Multistability for Global Property Tuning. (submitted).
2. **Sim, J.,** Lu, L., & Zhao, R. R., Electromagnetic (EM)-Driven Functional Materials. *Advanced Materials.* e21268 (2026).
3. **Sim, J.**, Wu, S., Hwang, S., Lu, L., & Zhao, R. R., Selective Actuation Enabled Multifunctional Magneto-mechanical Metamaterial for Programming Elastic Wave Propagation. *Advanced Functional Materials*. 34(16), 2422325 (2025)
4. **Sim, J.** & Zhao, R. R., Magneto-Mechanical Metamaterials: A Perspective. *Journal of Applied Mechanics*. 91(3), 031004 (2024).
5. **Sim, J.**, Wu, S., Dai, J., & Zhao, R. R., Magneto-Mechanical Bilayer Metamaterial with Global Area-Preserving Density Tunability for Acoustic Wave Regulation. *Advanced Materials.* 35(35), 2303541 (2023). Cover Article.
6. Yang, X.*, **Sim, J.***, Huang, W., & Zhao, R. R., Bioinspired Synergistic Texture and Color Modulation Enabled by Surface Instability of Cholesteric Liquid Crystal Elastomers. *Science Advances*. 11(44), eaea9183 (2025).
7. Averitt, S., **Sim, J.**, & Zhao, R. R., Spatiotemporal programming of shape memory polymers for tunable vibration responses. *Nature Communications* (under revision, 2026).
8. Pavlick, W.D., **Sim, J.,** West, D.L., Zhao, R. R., Ghalichechian, N., Complex Permittivity and Permeability of NdFeB Doped Polymers. *Research* (under review, 2026).
9. Lee, D. D., Wang, L., Chen, B., Tsai, Y., **Sim, J.**, Zhao, R. R., Zhang, X. S., Chen, W., Co-Design of Programmable Material Systems. *npj Metamaterials*. 2(1), 29 (2026).
10. Lee, J., **Sim, J.**, Yang, X., Wu, S., Averitt, S., & Zhao, R. R., Porous Liquid Crystal Elastomers with Thermally Tunable Acoustic Properties. *Materials Today*. 98, 103394 (2026).
11. Yang, X., **Sim, J.**, & Zhao, R. R., Spatiotemporal modulation of surface texture for information encoding and object manipulation. *Nature Communications.* 17, 7817 (2026).
12. Wu, S.*, Chang, Y.*, Leanza, S.*, **Sim, J.**, Lu, L., Stone, D., & Zhao, R. R., Magnetic Milli-Spinner for Robotic Endovascular Surgery. *Advanced Materials*. 38(4), e08180 (2025). Cover Article.
13. Chang, Y., Li, G., **Sim, J.**, Karniadakis, G., & Zhao, R. R., Clot Treatment via Compression- and Shear-Induced Densification of Fibrin Network Microstructure: A Combined in Vitro and in Silico Investigation. *Extreme Mechanics Letters.* 102391 (2025).
14. Averitt, S., **Sim, J.**, & Zhao, R. R., AI and Computing for Active Metamaterial Design: A Perspective. *Journal of Applied Mechanics*. 92(9), 094001 (2025).
15. King, P., **Sim, J.**, Shahed, K., Harwood, C., & Manogharan, G. Hands-on digital tools for metal casting engineering education to increase student engagement in foundry science. *International Journal of Metalcasting*. 1-13 (2024).
16. Lu, L., **Sim, J.**, & Zhao, R. R., Mechanics of hard-magnetic soft materials. *Mechanics of Materials*. 189, 104874 (2024). Cover Article.

17. Arunachala, P. K., Abrari Vajari, S., Neuner, M., **Sim, J. S.**, Zhao, R., Linder, C., A multiscale anisotropic polymer network model coupled with phase field fracture. *International Journal for Numerical Methods in Engineering*. 125(13), e7488 (2024).
18. West, D., Pavlick, W., **Sim, J.**, Dai, J., Wu, S., Eichenberger, J., Zhao, R. R., Ghalichechian, N., Reconfiguration of Electromagnetic Metasurfaces Using Tunable Shape Morphing Structures. *2024 18th European Conference on Antennas and Propagation (EuCAP)*. (2024)
19. Martinez, D., King, P., Sama, S. R., **Sim, J.**, Toykoc, H., & Manogharan, G., Effect of freezing range on reducing casting defects through 3D sand-printed mold designs. *The International Journal of Advanced Manufacturing Technology*, 126(1-2), 569-581(2023).
20. Bate, C., King, P., **Sim, J.**, & Manogharan, G., A Novel Approach to Visualize Liquid Aluminum Flow to Advance Casting Science. *Materials*, 16(2), 756 (2023).

CONFERENCE PRESENTATIONS

1. **Sim, J.**, Hwang, S., Chang, K., Kim, C., & Zhao, R. R., Photothermal Spatiotemporal Stiffness Programming for Diverse Mechanical Responses, Oral presentation, ASME IMECE, Vancouver, Canada, November 8 – 12, 2026.
2. **Sim, J.**, Wu, S., Hwang, S., Lu, L., & Zhao, R. R. (2025). Selective Actuation Enabled Multifunctional Magneto-Mechanical Metamaterial for Programming Elastic Wave Propagation, Oral presentation, Society of Engineering Science Annual Technical Meeting, Atlanta, GA, October 12-15, 2025.
3. **Sim, J.**, Wu, S., Dai, J., & Zhao, R. R. (2023). Magneto-Mechanical Bilayer Metamaterial with Global Area-Preserving Density Tunability for Acoustic Wave Regulation, Oral presentation, Stanford Mechanical Engineering Conference, Stanford, CA, May 9, 2024.
4. **Sim, J.**, Wu, S., Dai, J., & Zhao, R. R. (2023). Magneto-Mechanical Bilayer Metamaterial with Global Area-Preserving Density Tunability for Acoustic Wave Regulation, Poster session, Stanford Mechanical Engineering Conference, Stanford, CA, May 9, 2024.
5. **Sim, J.**, Wu, S., Dai, J., & Zhao, R. R. (2023). Magneto-Mechanical Bilayer Metamaterial with Global Area-Preserving Density Tunability for Acoustic Wave Regulation, Oral presentation, Society of Engineering Science Annual Technical Meeting, Minneapolis, MN, October 8-11, 2023.
6. **Sim, J.**, Wu, S., Dai, J., & Zhao, R. R. (2023). Area-Preserving Magneto-Mechanical Bilayer Metasurface for Acoustic Wave Regulation, Poster session, Society of Engineering Science Annual Technical Meeting, Minneapolis, MN, October 8-11, 2023.

PROFESSIONAL ACTIVITIES

Technical Committee Service

1. Secretary, Materials for Biomimetic and Medical Applications Technical Committee, Materials Division, ASME 2026-2027

Conference Symposium Organizer and Session Chair

1. Co-Organizer, Berkeley/Stanford Computational Mechanics Festival (CompFest), January 30, 2027. **Website**.
2. Session chair, Functional Soft Composites – Design, Mechanics, and Manufacturing, ASME IMECE, Vancouver, BC, Canada, November 8-12, 2026.

3. Organizer, Track 13: Mechanics of Solids, Structures, and Fluids Topics: Functional Soft Composites – Design, Mechanics, and Manufacturing, ASME IMECE, Vancouver, BC, Canada, November 8-12, 2026.
4. Session chair, Symposium: 6.1 Functional Soft Composites – Design, Mechanics, and Manufacturing. Society of Engineering Science Annual Technical Meeting, Minneapolis, MN, October 8-11, 2023.

Departmental Service

1. Student Committee Member, Faculty Search Committee (2025–2026), Department of Mechanical Engineering, Stanford University, February – April 2026.

Journal Reviews

Small Methods; SmartBot; Communications Engineering; Materials & Design

Professional Societies

American Society of Mechanical Engineering (ASME)

Society of Engineering Science (SES)

Materials Research Society (MRS)

TEACHING

Course Assistant and Lab Instructor
ME 303: Soft Composites and Soft Robotics
Fall 2023, 2024, 2025, Spring 2025
Department of Mechanical Engineering, Stanford University

- Designed lab sessions (3 total per quarter) and taught experimental fabrication and characterization of soft materials, soft electronic sensors, and shape-morphing structures.
- Held weekly office hours, lab sessions, graded assignments, answered online discussion board and gave ABAQUS finite element analysis lecture.
- Class size: 60 students. Teaching evaluations available on request. **Final projects**.

OUTREACH

Stanford seeME Instructor Winter - Spring 2026
Stanford Department of Mechanical Engineering, Stanford University

- Designed a class on origami applications in mechanical engineering for local high school and middle school students at an annual on-campus outreach event.

Summer Undergraduate Research Fellowship (SURF) Mentor Summers 2024, 2025
Stanford College of Engineering, Stanford University
The program has a strong focus on training underrepresented minority or first-generation undergraduate student engineers who can promote diversity in STEM.

- Mentored an undergraduate student from Tufts University on the basics of experimental research and technical communication. The student was trained to conduct research and present an end of program poster session.

NSF REU Mentor Summers 2023 – 2026
Stanford Undergraduate Research Institute (SURI), Stanford University

- Mentored and trained multiple Stanford University mechanical engineering undergraduate students in general mechanical engineering research practices and helped prepare students for the end of program poster session.
- Fulfilled this role across multiple summer academic quarters at Stanford University.

Summer Research Program Mentor June 2023 – July 2023
Stanford Robotics Center, Stanford University

- Mentored two local Bay Area high school students in STEM concepts in a six-week program designed to promote robotics and engineering education in first generation, minoritized, and underrepresented students.

MENTORING

Munashe Chibinyani	Summer 2026 – present	Ph.D. Rotation, Mechanical Engineering, Stanford University
Chloe Kim	2025 – present	BS Mechanical Engineering, Stanford University
Sarah Hwang	2023 – present	BS/MS Mechanical Engineering, Stanford University
Kendra Chang	Summer 2026	BS Mechanical Engineering, Stanford University
Yizhi Chen	Summer 2026	BS Mechanical Engineering, University of Michigan
Yufei Chen	Summer 2026	BS Computer Science, Stanford University
Joseph Jimenez	Spring 2026	BS Mechanical Engineering, Stanford University
Ellia Cavanaugh	Spring 2026	BS Mechanical Engineering, Stanford University
Chisa Ogaki	2025 – 2026	BS Mechanical Engineering, Stanford University
Samantha Averitt	Fall 2025	Ph.D. Rotation, Mechanical Engineering, Stanford University
Ariana Olivares	2024 – 2025	BS Mechanical Engineering, Tufts University
Caleb Nangle	Winter 2025	BS Mechanical Engineering, Stanford University
Phoenix Clarke	Summer 2024	BS Mechanical Engineering, Stanford University
Annika Yong	2023 – 2024	MS Mechanical Engineering, Stanford University
Eli Jones	Spring 2023	BS Mechanical Engineering, Stanford University

3 high school students mentored, names omitted for privacy